Force Systems

- FORCE SYSTEMS
- MOMENTS OR TORQUE
- COUPLE MOMENT

Force Systems

VIDEO

[1]

[1] – Youtube accessed on February 22, 2015 (https://www.youtube.com/watch?v=MfNz7DRwtQQ)

Force Systems

Newton's Laws of Motion

What is Newton's First Law of Motion?

A body at rest remains at rest and a body in motion continues its motion in a straight line unless acted upon by an external force

What is Newton's Second Law of Motion?

The force action on a body gives it an acceleration which is in the direction of force

$$\boldsymbol{F = m\, a}$$

What is Newton's Third Law of Motion?

If one object exerts a force on another object, the other exerts the same force in the opposite direction on the one

Action = - Reaction

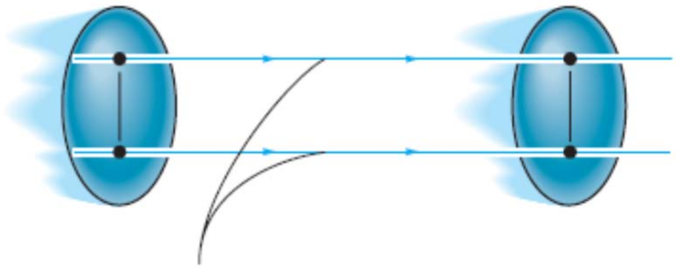
Contact force

Force Systems

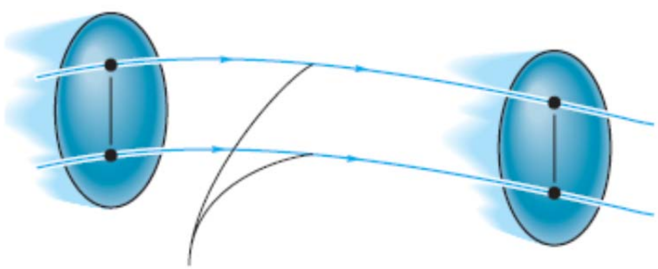
FORCE & MOTION

What kind of motion can a force generate in a rigid body?

Translation



Path of rectilinear translation



Path of curvilinear translation [1]

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp. 312

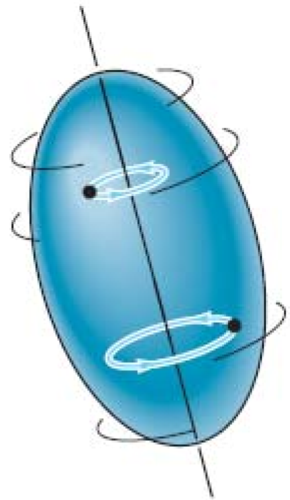
Force Systems

Force & Motion

What kind of motion can a force generate in a rigid body?

Translation

Rotation about a fixed axis



Rotation about a fixed axis [1]

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp. 312

Force Systems

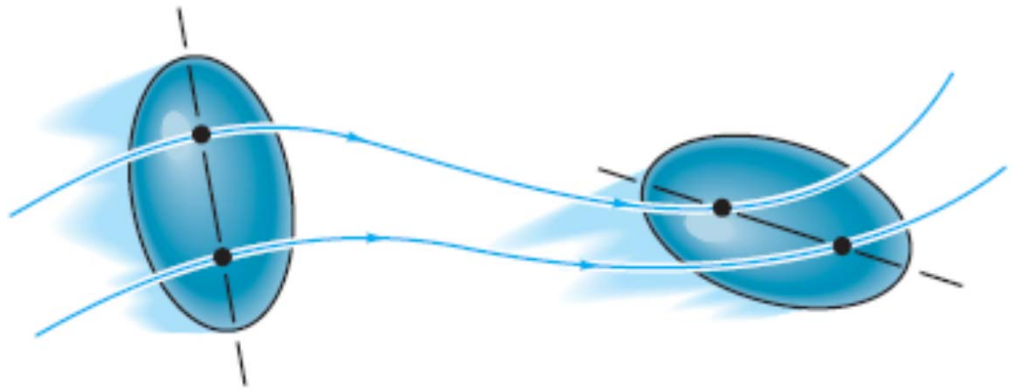
Force & Motion

What kind of motion can a force generate in a rigid body?

Translation

Rotation about a fixed axis

General Motion



[1]

General plane motion

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp. 312

Force Systems

MOMENTS OR TORQUE

What is the tendency of a force to produce rotary motion in a body called?

Moment or Torque

What are the factors on which the magnitude of moment depends?

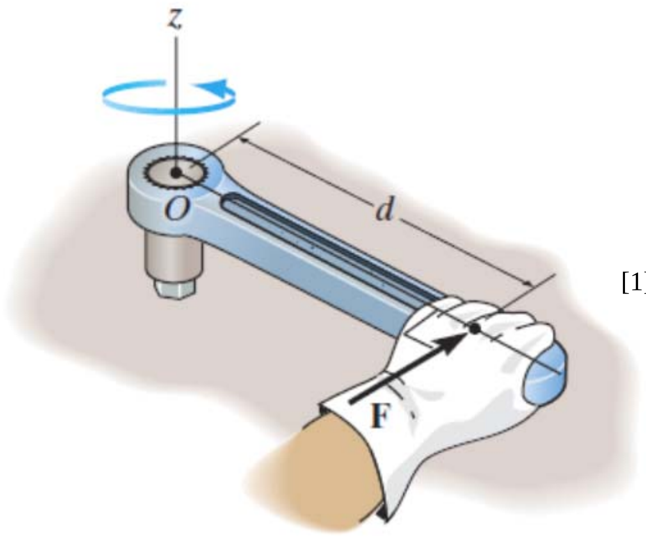
Magnitude of force, F

Perpendicular distance from axis of rotation d

$$M = F\,d$$

The magnitude of the moment is directly proportional to the magnitude of Force F and the <u>perpendicular</u> distance or moment arm d

Moment is defined with respect to a fix reference point



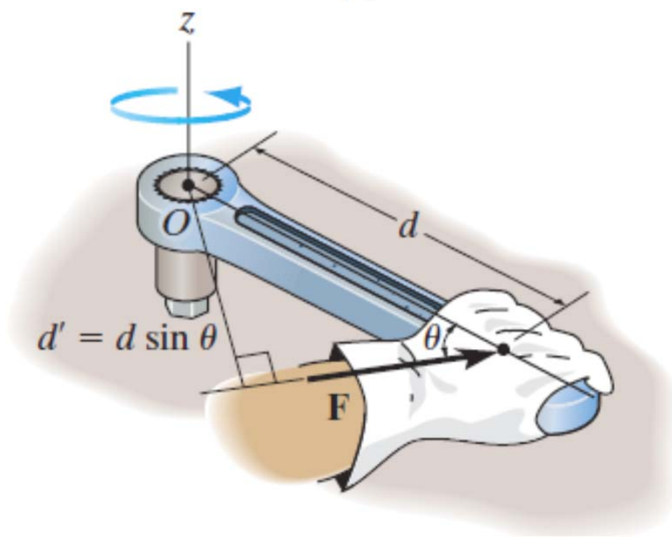
[1]

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp. 117

Moments or Torque

Basic Concepts

What if the force is applied at an angle?



[1]

The component of distance is taken, which is perpendicular to the force

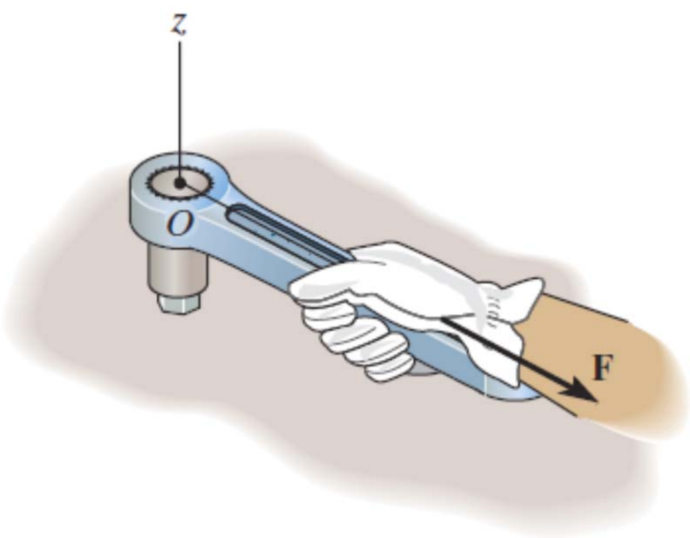
Or the component of force is taken, which is perpendicular to the distance

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp. 117

Moments or Torque

BASIC CONCEPTS

What if the force is passing through the reference point?



[1]

No Moment

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp. 117

Moments or Torque

Vector Notation

Is moment a scalar or a vector?

Vector

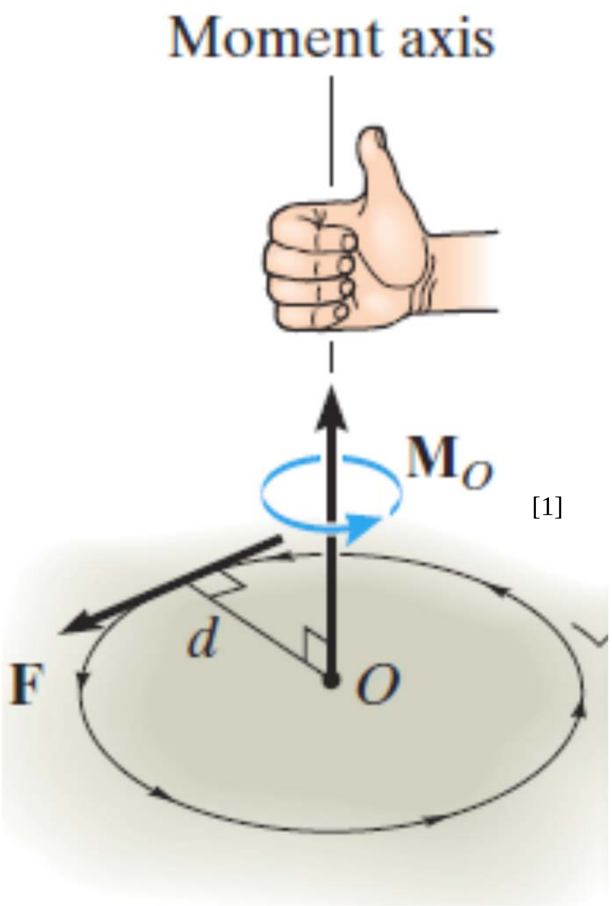
What about its direction?

It is perpendicular to the plane that contains the force and its moment arm

Which mathematical operator, when applied to two vectors, give an other vector which is perpendicular to both?

$$\underline{\boldsymbol{M_o}} = \underline{\boldsymbol{r}} \times \underline{\boldsymbol{F}}$$

$\boldsymbol{r}$ can be any position vector, measured from point O to any point on the line of action of the force F



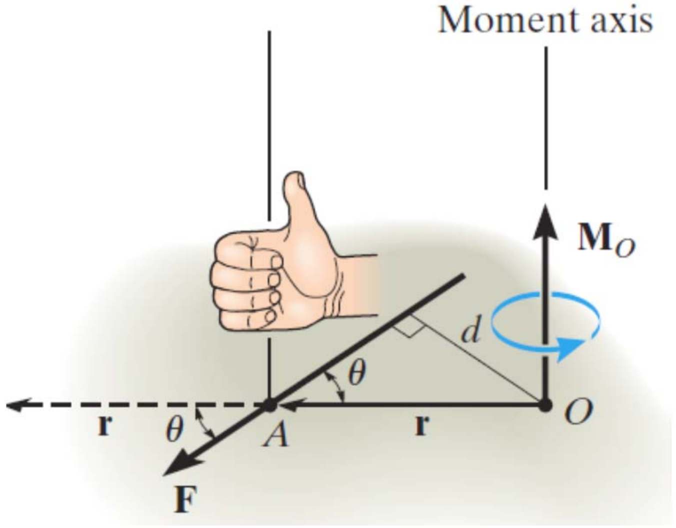
[1]

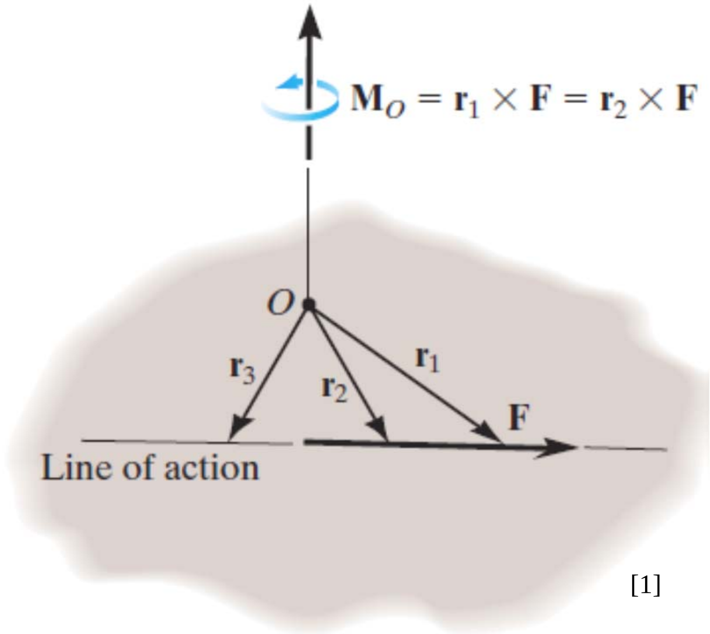
[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp. 118

Moments or Torque

PRINCIPLE OF TRANSMISSIBILITY

$$\underline{M_o} = \underline{r} \times \underline{F}$$





[1]

F can be applied at any point along its line of action and still create this same moment about point O

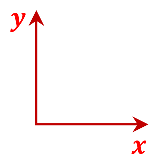
This property is called the principle of transmissibility of a force

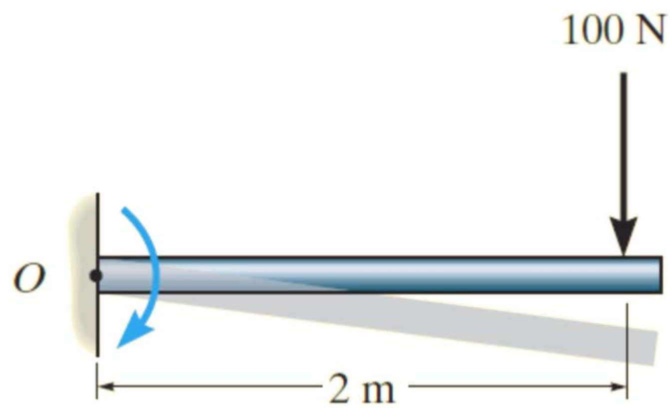
[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp. 118

Moments or Torque

EXAMPLE I

Calculate moment about point O





Magnitude:

$$M_o = F\,d = (100)(2) = 200\ Nm$$

Direction:

-ve z-axis

$$\underline{r} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} m \qquad \underline{F} = \begin{bmatrix} 0 \\ -100 \\ 0 \end{bmatrix} N$$

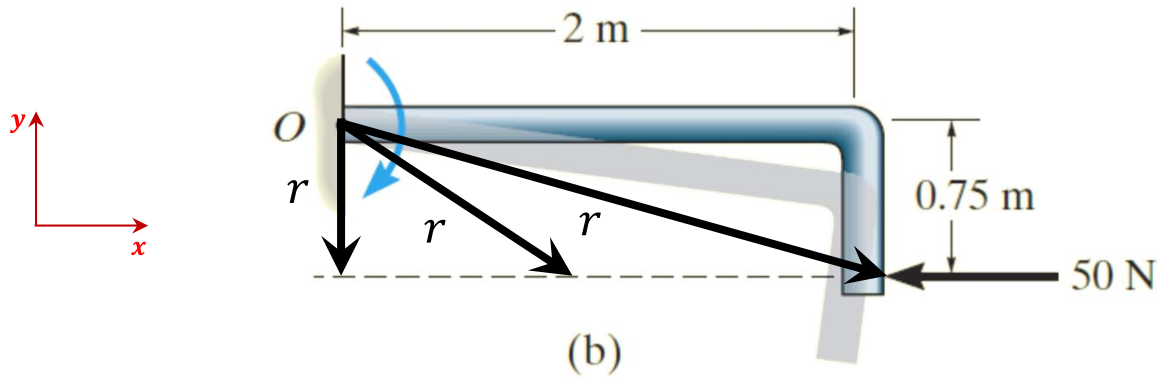
$$\underline{M_o} = \underline{r} \times \underline{F}$$

$$\underline{M_o} = \begin{bmatrix} 0 \\ 0 \\ -200 \end{bmatrix} Nm$$

Moments or Torque

Example II

Calculate moment about point O



(b)

Magnitude:

$$M_o = F\, d = (50)(0.75) = 37.5\ Nm$$

Direction:

-ve z-axis

$$\underline{r} = \begin{bmatrix} 2 \\ -0.75 \\ 0 \end{bmatrix} m \qquad \underline{F} = \begin{bmatrix} -50 \\ 0 \\ 0 \end{bmatrix} N$$

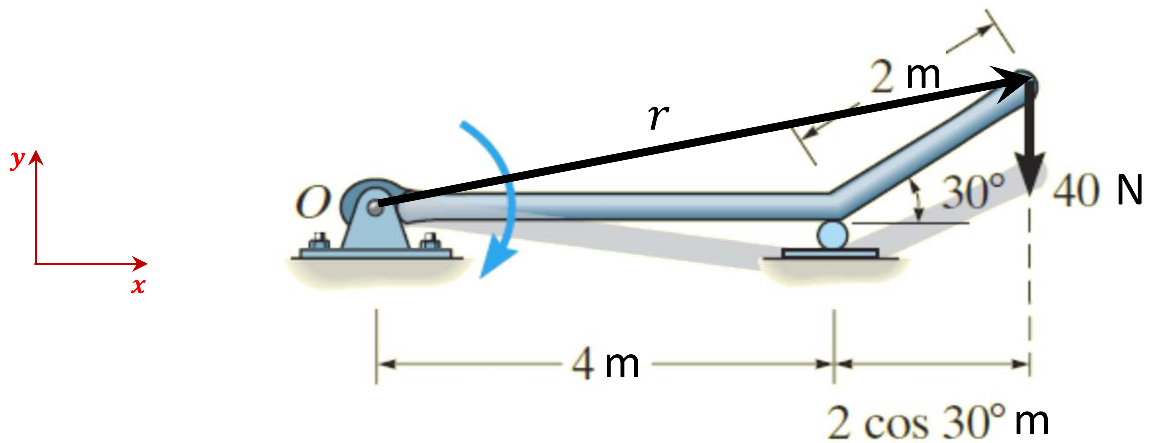
$$\underline{M_o} = \underline{r} \times \underline{F}$$

$$\underline{M_o} = \begin{bmatrix} 0 \\ 0 \\ -37.5 \end{bmatrix} Nm$$

Moments or Torque

Example III

Calculate moment about point O



Magnitude:

$$M_o = F\,d = (40)(4 + 2cos(30°)) = 229.282Nm$$

Direction:

-ve z-axis

$$\underline{r} = \begin{bmatrix} 5.732 \\ 1 \\ 0 \end{bmatrix} m \qquad \underline{F} = \begin{bmatrix} 0 \\ -40 \\ 0 \end{bmatrix} N$$

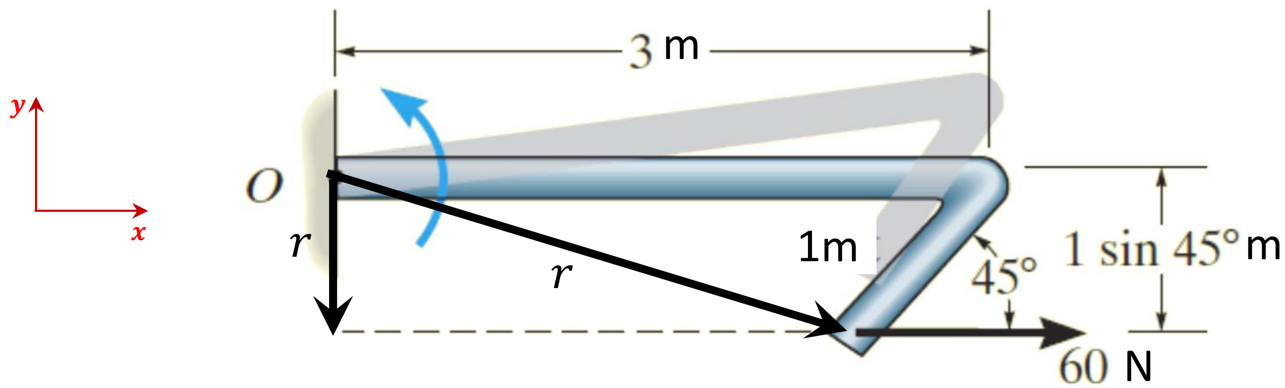
$$\underline{M_o} = \underline{r} \times \underline{F}$$

$$\underline{M_o} = \begin{bmatrix} 0 \\ 0 \\ -229.283 \end{bmatrix} Nm$$

Moments or Torque

Example IV

Calculate moment about point O



Magnitude:

$$M_o = F\,d = (60)(1\,sin(45°)) = 42.426Nm$$

Direction:

+ve z-axis

$$\underline{r} = \begin{bmatrix} 3 - cos\,45° \\ sin\,45° \\ 0 \end{bmatrix} m \qquad \underline{F} = \begin{bmatrix} 60 \\ 0 \\ 0 \end{bmatrix} N$$

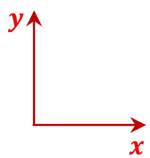
$$\underline{M_o} = \underline{r} \times \underline{F}$$

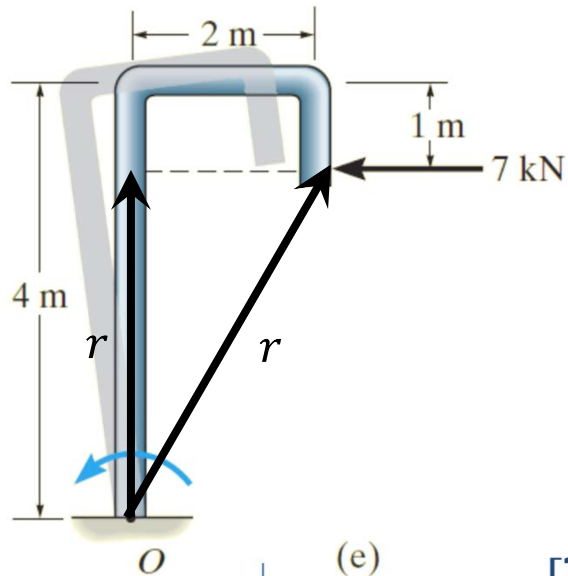
$$\underline{M_o} = \begin{bmatrix} 0 \\ 0 \\ 42.426 \end{bmatrix} Nm$$

Moments or Torque

Example V

Calculate moment about point O





(e)

Magnitude:

$$M_o = F\,d = (7000)(4-1) = 21kNm$$

Direction:

+ z-axis

$$\underline{r} = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} m \qquad \underline{F} = \begin{bmatrix} -7000 \\ 0 \\ 0 \end{bmatrix} N$$

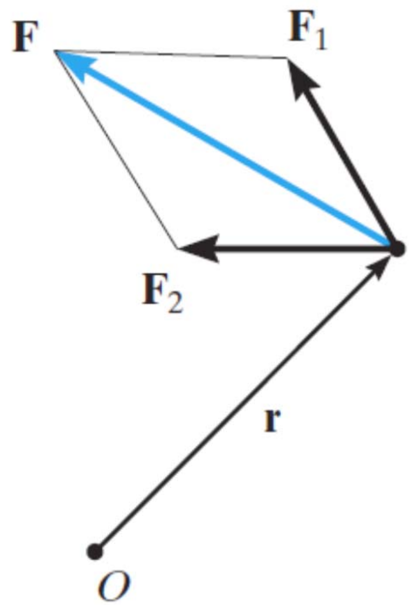
$$\underline{M_o} = \underline{r} \times \underline{F}$$

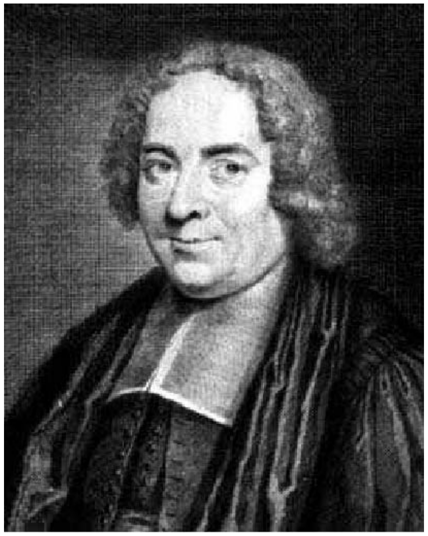
$$\underline{M_o} = \begin{bmatrix} 0 \\ 0 \\ 21000 \end{bmatrix} Nm$$

Moments or Torque

PRINCIPLE OF MOMENTS OR VARIGNON'S THEOREM

the moment of a force about a point is equal to the sum of the moments of the components of the force about the point





Pierre Varignon (1654-1722)

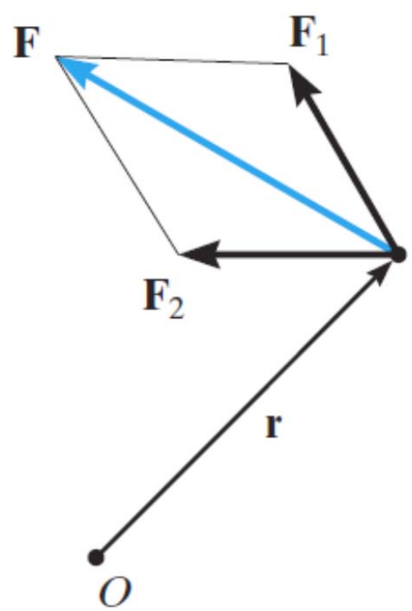
$$\underline{\boldsymbol{M_0}} = \underline{\boldsymbol{r}} \times \underline{\boldsymbol{F}} = \underline{\boldsymbol{r}}\,(\underline{\boldsymbol{F_1}} + \underline{\boldsymbol{F_2}})$$

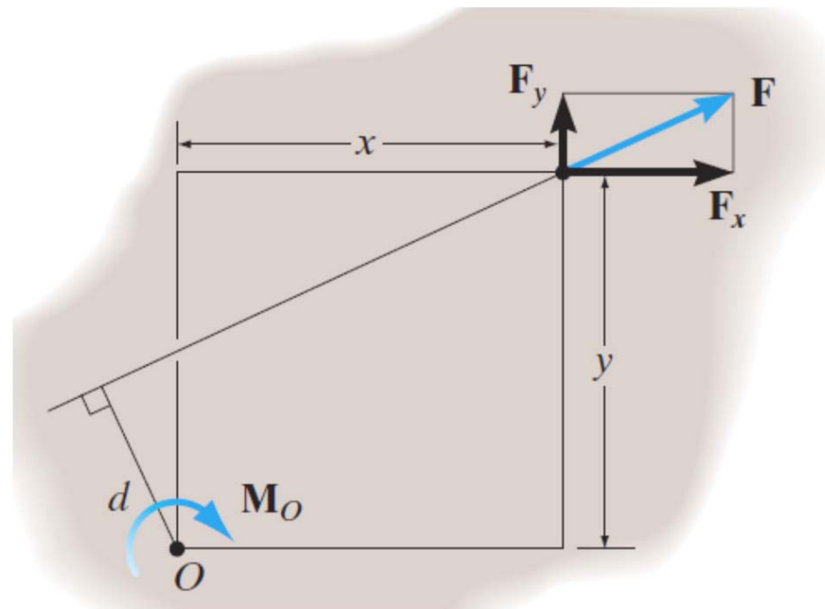
$$\underline{\boldsymbol{M_0}} = \underline{\boldsymbol{r}} \times \underline{\boldsymbol{F}} = \underline{\boldsymbol{r}} \times \underline{\boldsymbol{F_1}} + \underline{\boldsymbol{r}} \times \underline{\boldsymbol{F_2}}$$

Moments or Torque

PRINCIPLE OF MOMENTS OR VARIGNON'S THEOREM

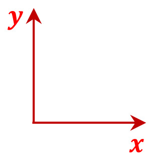
the moment of a force about a point is equal to the sum of the moments of the components of the force about the point

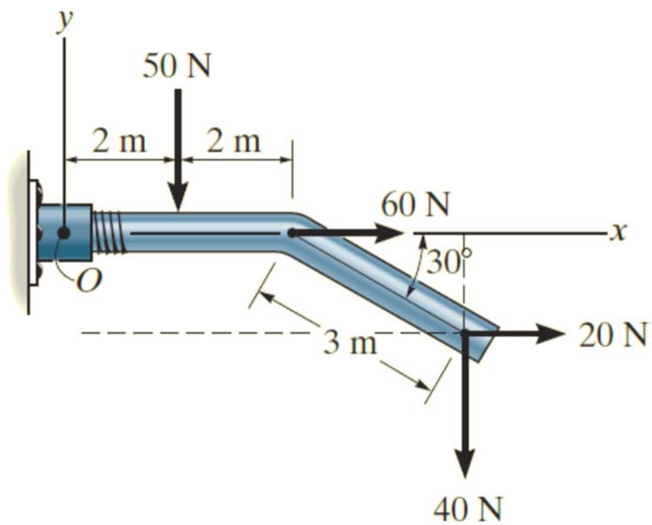




$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} = \mathbf{r} \times (\mathbf{F}_1 + \mathbf{F}_2) = \mathbf{r} \times \mathbf{F}_1 + \mathbf{r} \times \mathbf{F}_2$$

Calculate the resultant moment about point O





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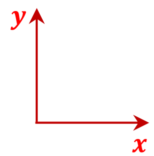
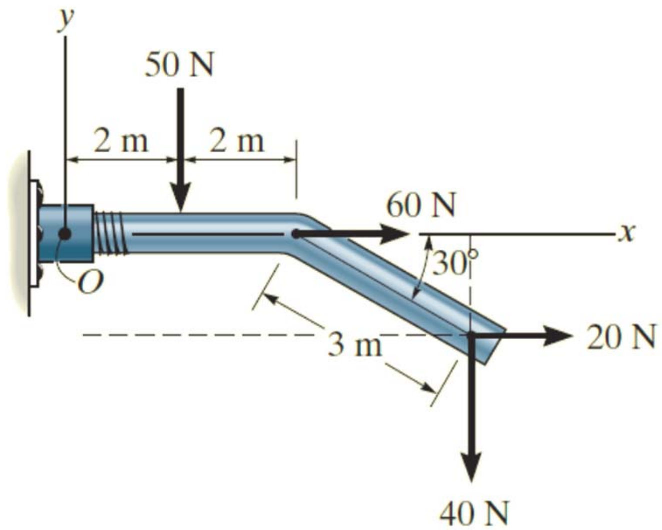
Moments or Torque

EXAMPLE VI

Calculate the resultant moment about point O





Formulation:

$F_1 = 50N, F_2 = 60N, F_3 = 20N, F_4 = 40N$

$d_1 = 2m, d_2 = 0m, d_3 = 3\ sin\ 30°\ m$,
$d_4 = 4 + 3\ cos\ 30°\ m$

Magnitude:

$$M_o = F_1\, d_1 + F_2\, d_2 + F_3\, d_3 + F_4\, d_4$$
$$= -(50)(2) + 60(0) + 20(3\ sin\ 30°)$$
$$-(40)(4 + 3\ cos\ 30°) = -333.923\ Nm$$

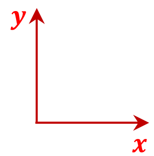
Direction:

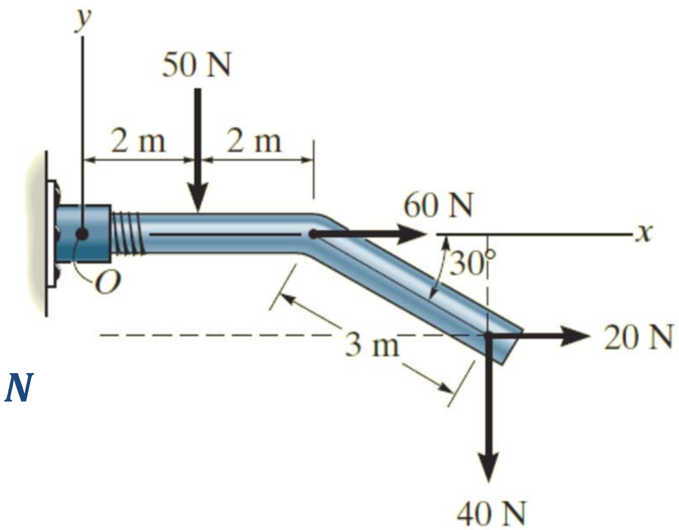
-ve z-axis

Moments or Torque

Example VI

Calculate the resultant moment about point O





Formulation:

$$\underline{r_1} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} m \qquad \underline{F_1} = \begin{bmatrix} 0 \\ -50 \\ 0 \end{bmatrix} N$$

$$\underline{r_2} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} m \qquad \underline{F_2} = \begin{bmatrix} 60 \\ 0 \\ 0 \end{bmatrix} N$$

$$\underline{r_3} = \begin{bmatrix} 4 + 3\cos 30° \\ -3\sin 30° \\ 0 \end{bmatrix} m \qquad \underline{F_3} = \begin{bmatrix} 20 \\ 0 \\ 0 \end{bmatrix} N$$

$$\underline{r_4} = \begin{bmatrix} 4 + 3\cos 30° \\ -3\sin 30° \\ 0 \end{bmatrix} m \qquad \underline{F_4} = \begin{bmatrix} 0 \\ -40 \\ 0 \end{bmatrix} N$$

Solution:

$$\underline{M_r} = \underline{M_1} + \underline{M_2} + \underline{M_3} + \underline{M_4}$$

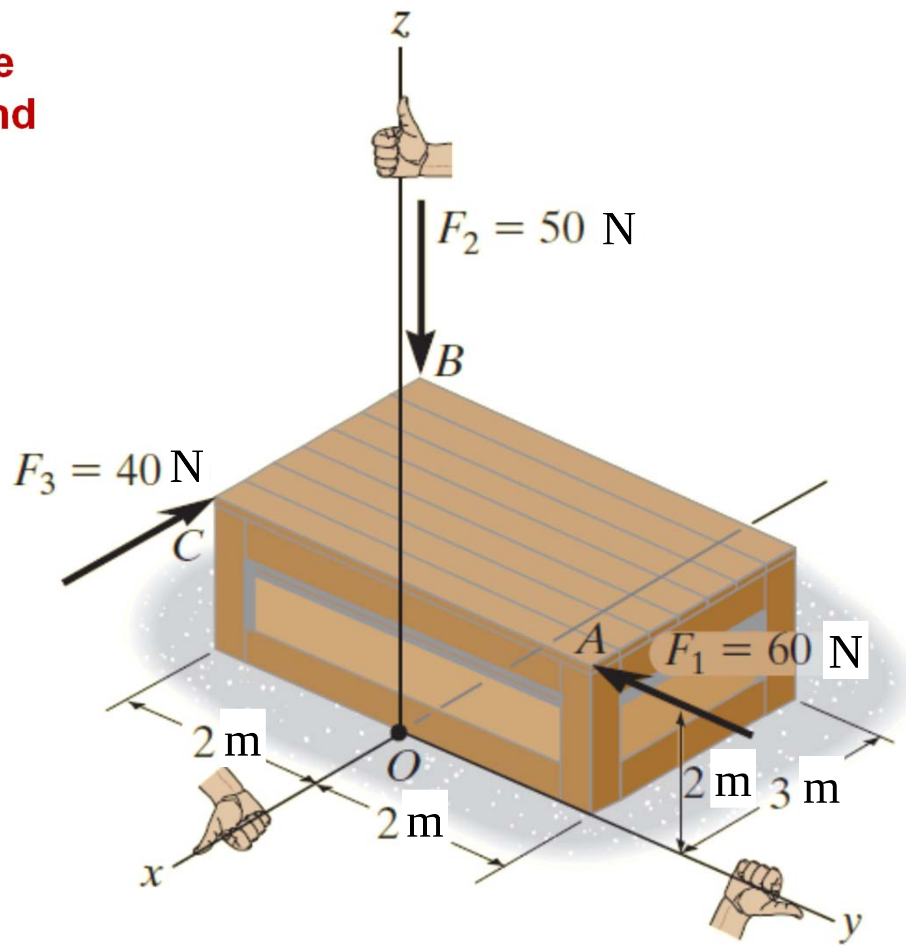
$$\underline{M_r} = \underline{r_1} \times \underline{F_1} + \underline{r_2} \times \underline{F_2} + \underline{r_3} \times \underline{F_3} + \underline{r_4} \times \underline{F_4}$$

$$\underline{M_r} = \begin{bmatrix} 0 \\ 0 \\ -333.923 \end{bmatrix} Nm$$

Moments or Torque

ABOUT A SPECIFIED AXIS

Determine the resultant moment of the three forces about x-axis, the y-axis, and the z-axis.



Moments or Torque

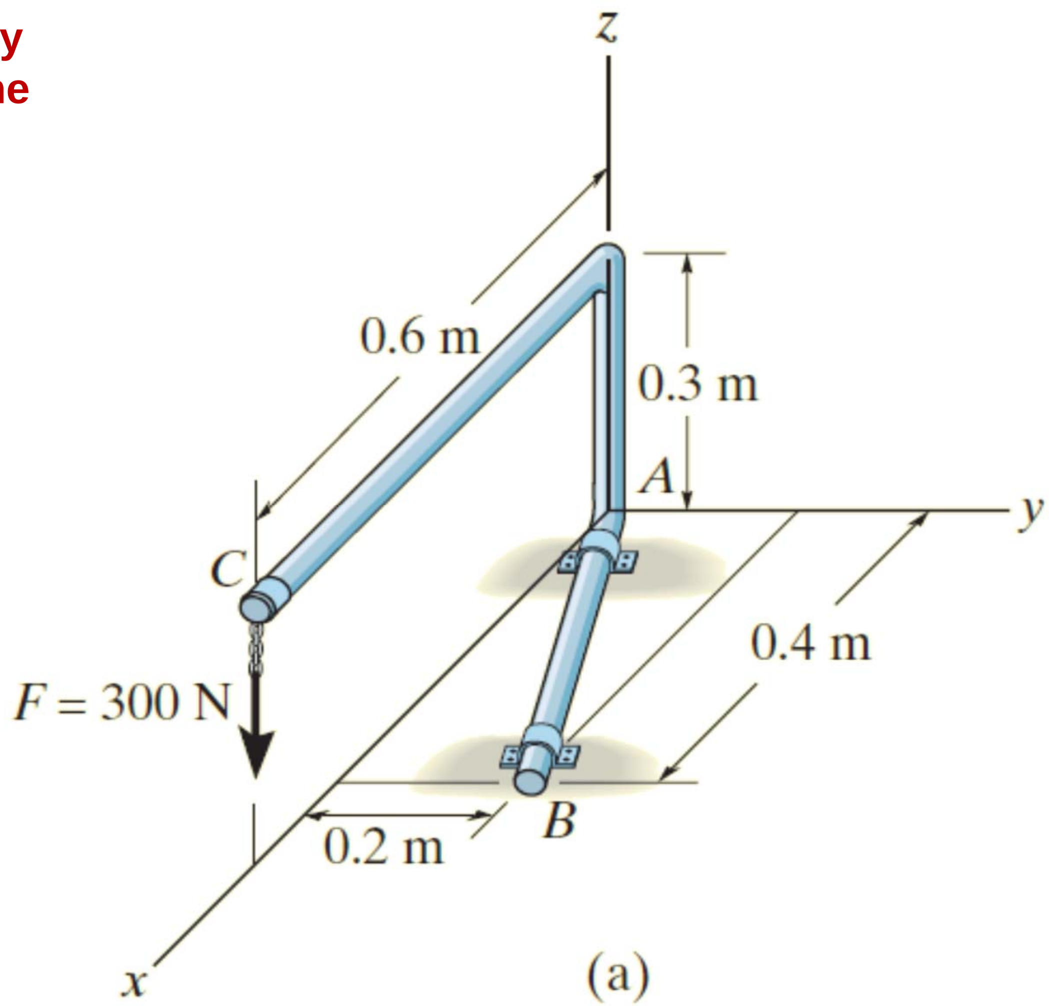
Example VII

Determine the moment produced by the force F, which tends to rotate the rod about the AB axis.

$$\boldsymbol{F} = \begin{bmatrix} 0 \\ 0 \\ -300 \end{bmatrix} N \qquad \underline{\boldsymbol{r_C}} = \begin{bmatrix} 0.6 \\ 0 \\ 0.3 \end{bmatrix} m$$

$$\underline{\boldsymbol{AB}} = \begin{bmatrix} 0.4 \\ 0.2 \\ 0 \end{bmatrix} m \qquad \underline{\boldsymbol{M_{AB}}} = ?$$

Finding the moment about A and then taking its projection along AB



Moments or Torque

EXAMPLE VII

$$\underline{M_o} = \underline{r_C} \times \underline{F}$$

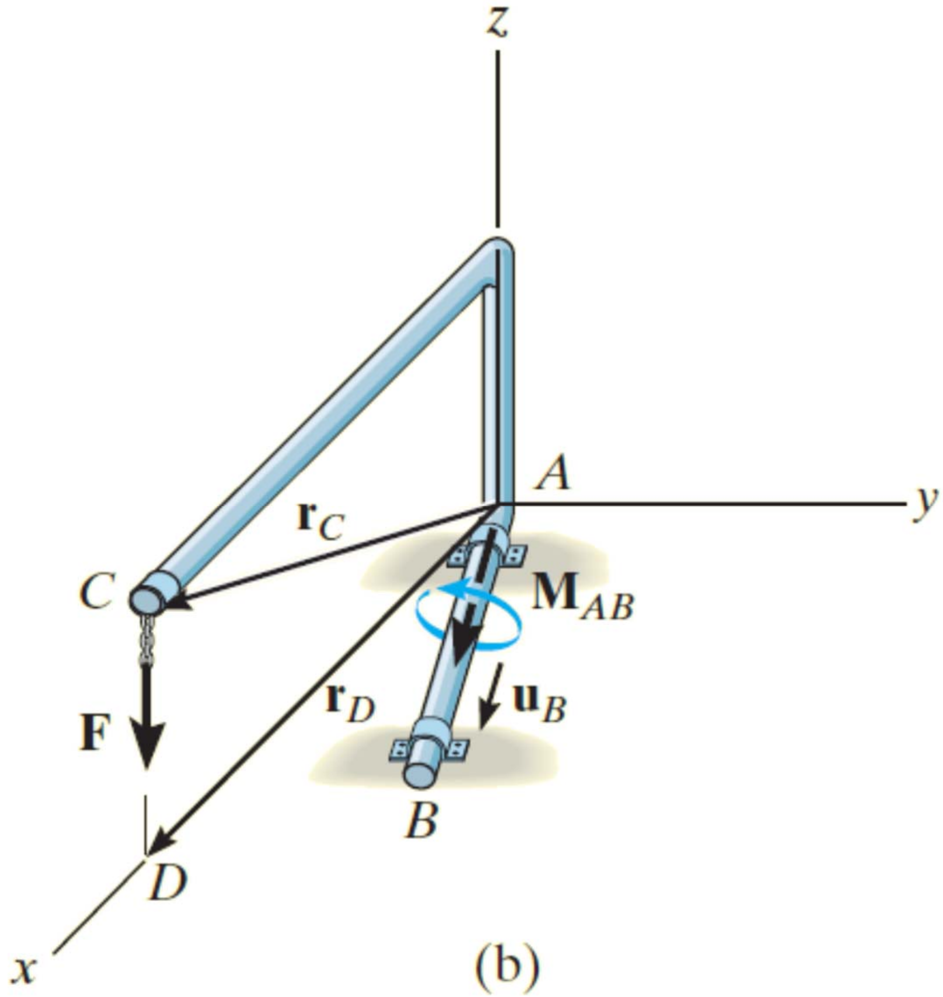
$$\underline{M_o} = \begin{bmatrix} 0.6 \\ 0 \\ 0.3 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ -300 \end{bmatrix}$$

$$\underline{M_o} = \begin{bmatrix} 0 \\ 180 \\ 0 \end{bmatrix} Nm$$

$$\underline{M_{AB}} = \underline{M_0} \,.\, \widehat{AB}$$

$$\underline{M_{AB}} = \begin{bmatrix} 0 \\ 180 \\ 0 \end{bmatrix} . \begin{bmatrix} 0.894 \\ 0.447 \\ 0 \end{bmatrix}$$

$$\underline{M_{AB}} = 80.498\ Nm$$



(b)

Couple Moment

COUPLE MOMENT

Calculate the resultant moment

$$M_o = (30)(0.2) + (30)(0.2)$$

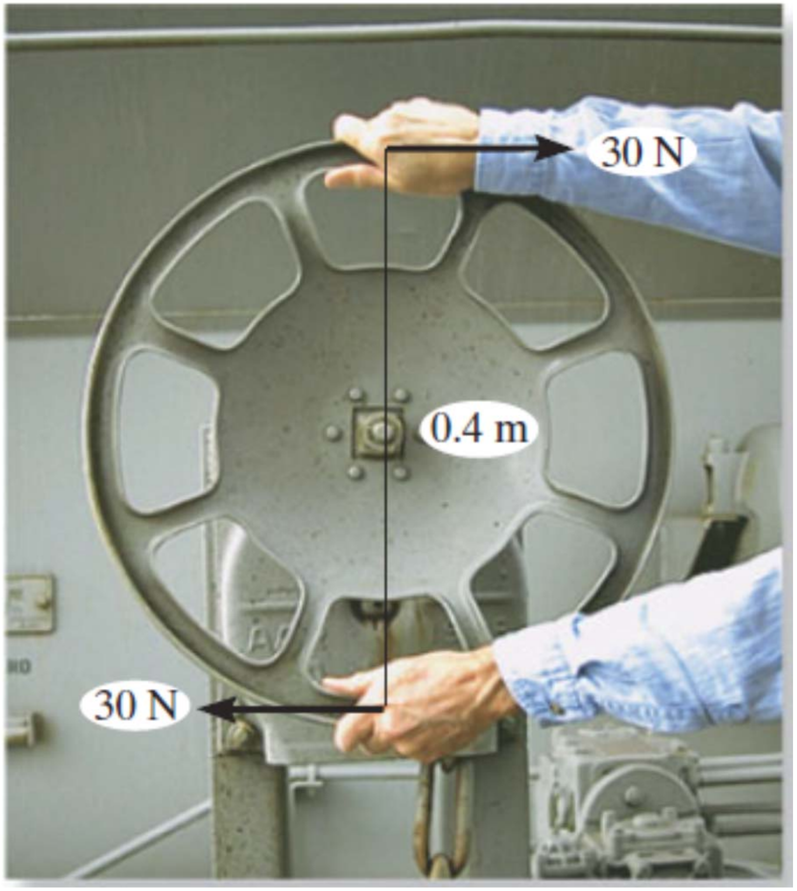
$$M_o = (30)(0.2 + 0.2)$$

$$M_o = (30)(0.4)$$

$$M_o = 12\ Nm$$

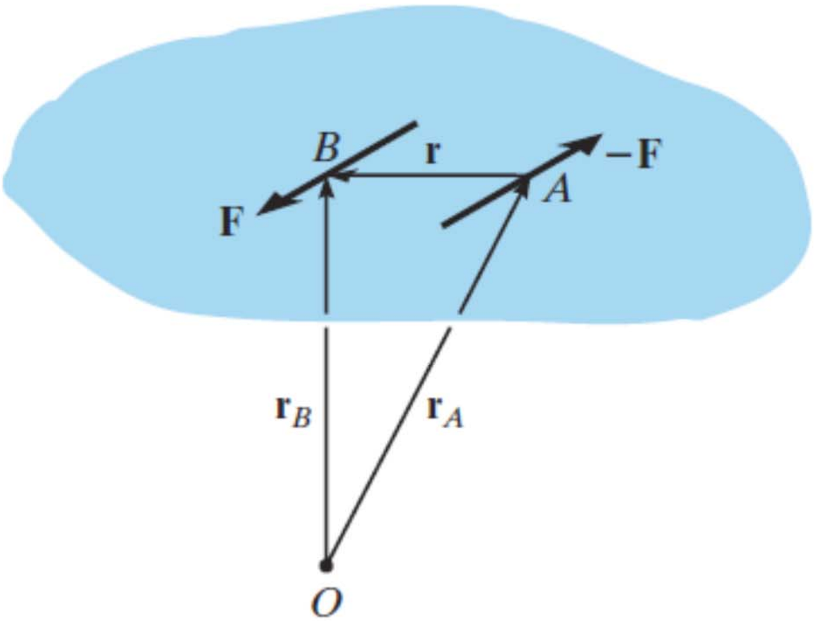
What generates a couple moment?

Two parallel forces that have the same magnitude, but opposite directions, and are separated by a perpendicular distance



Couple Moment

Couple Moment



$$\mathbf{M} = \mathbf{r}_B \times \mathbf{F} + \mathbf{r}_A \times -\mathbf{F} = (\mathbf{r}_B - \mathbf{r}_A) \times \mathbf{F}$$

$$\mathbf{r} = \mathbf{r}_B - \mathbf{r}_A.$$

$$\mathbf{M} = \mathbf{r} \times \mathbf{F}$$

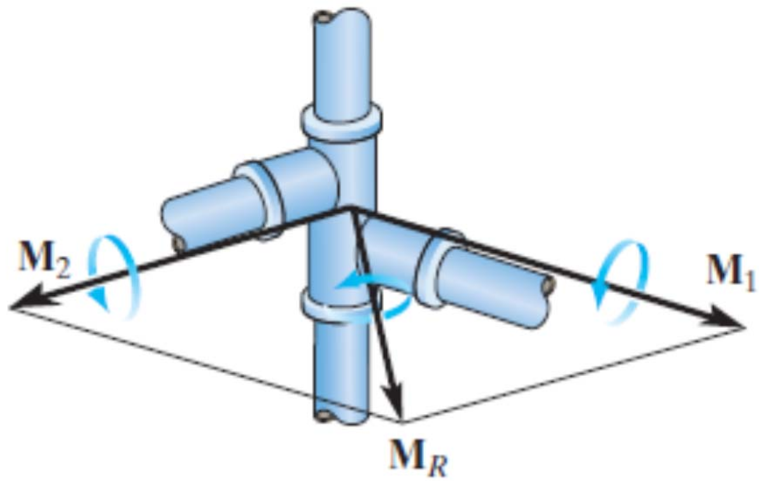
A couple moment is a free vector

It can act at any point and would result the same

Couple Moment

Resultant

resultant couple moment is simply the vector sum of all the couple moments of the system.



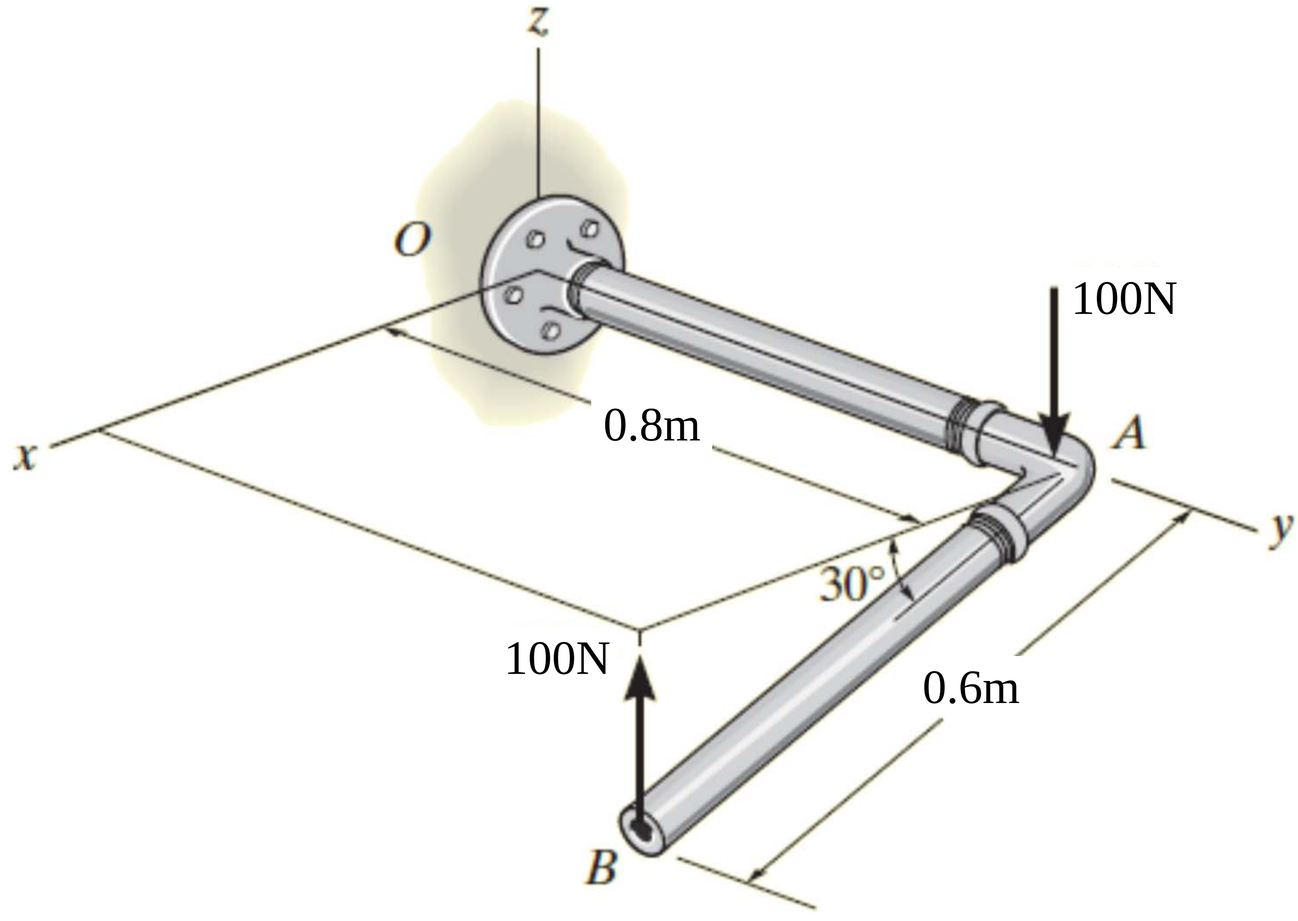
$$\mathbf{M}_R = \mathbf{M}_1 + \mathbf{M}_2$$

$$\mathbf{M}_R = \Sigma(\mathbf{r} \times \mathbf{F})$$

Couple Moment

Example VIII

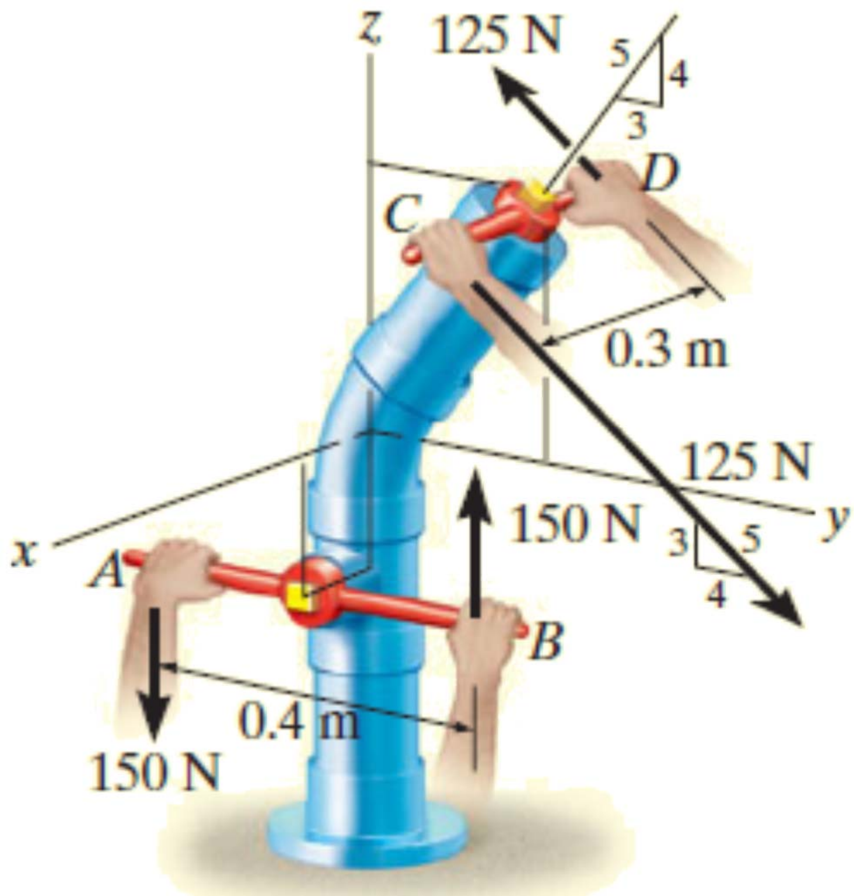
Determine the moment acting on the pipe



Couple Moment

EXAMPLE IX

Determine the moment acting on the pipe



Couple Moment

Example IX

Determine the moment acting on the pipe